Document Change Check – Proof of Concept Overview

Prototype findings and feasibility outlook for automatically verifying Data House updates

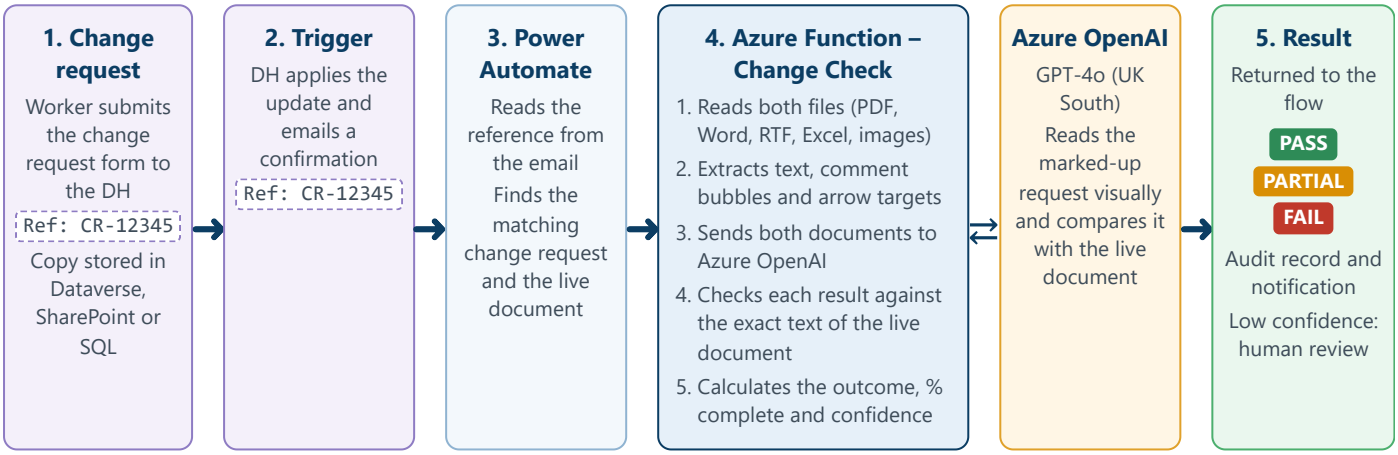
Draft – 9 October 2026

Purpose

A worker submits a change request form, carrying a unique reference, to the **Data House (DH)**. When the DH has applied the update, it sends an email confirming the update and quoting that reference. The email starts a Power Automate flow, which uses the reference to find the original change request and sends it with the updated (live) document to an Azure Function. The function's **Change Check** method uses Azure OpenAI (GPT-4o, which can read images) to find every requested change, then checks whether each one appears in the live document. It returns an overall outcome and a line-by-line breakdown of each change.

This replaces a manual side-by-side comparison of the two documents. Every DH confirmation is independently checked within moments of arriving, giving a consistent, auditable answer and highlighting any changes that still need attention.

How it works



The flow runs automatically for every DH confirmation email. Results with low confidence are routed to a person for human-in-the-loop review.

PASS Every requested change has been applied correctly.

PARTIAL At least one change has been applied and at least one has not.

FAIL None of the requested changes have been applied correctly.

Capabilities

- **Any common format:** PDF, Word (.docx), RTF, Excel, PowerPoint, text and images (PNG, JPG).
- **Reads mark-ups:** comment bubbles, arrows, text boxes, handwriting and "should be" notes, across every page.
- **Counts only explicit requests:** printed or unchanged fields are not treated as changes, and red or black redaction bars are ignored.
- **Handles each kind of edit:** replacements, additions (for example, appending words to a sentence) and removals.
- **Per-change detail:** field, old value, new value, expected text, result and a note explaining any failure.
- **Exact matching:** spacing, spelling and punctuation must match ("AR1 2AO" is not "AR12AO", and "United Kingdon" fails), and the value must be in the right place, not elsewhere in the document.
- **Vague requests flagged:** comments such as "change this" are reported as *Unidentified Field Changes* for a person to review.
- **Consistent outcome:** the outcome and the % of changes completed are calculated by code, not by the AI.
- **Human in the loop:** each result has a 0–100 confidence score with a reason, lowered automatically when the code corrects the AI. Low-confidence and unidentified results are sent to a person for review.
- **Document reference:** the policy or reference number is taken from the document or file name ("Not Found" if there is none).
- **Cost visibility:** each run reports the model used, the tokens used and an estimated cost.
- **Secure:** calls require a function key, the AI key is stored only in Azure settings, and data is processed in UK South.

Linking the DH confirmation to the change request

Each DH confirmation email must be matched to the change request it relates to. This is done with a **unique change reference**:

- **Raised once:** each change request form is given a unique reference (for example `CR-12345`) when it is submitted, and a copy is stored under that reference in Dataverse, SharePoint or SQL.
- **Quoted back by the DH:** the DH includes the same reference in the confirmation email, ideally in a fixed format in the subject line.
- **Matched by the flow:** the flow reads the reference from the email, finds the matching change request and the live document, then runs the Change Check.
- **Exceptions handled:** if the reference is missing or doesn't match a request, the flow records an exception and alerts the team for manual review instead of running the check.
- **Double-checked:** the Change Check also reports the policy or document reference (docRef) it finds, which the flow can compare with the request as a second safeguard.

The reference format, and how the DH quotes it and supplies the updated document (as an email attachment or in a shared location), need to be agreed with the DH before build.

Example result

A motor policy schedule with three requested changes, of which one was applied correctly:

Field	Old value	New value	Live value	Result	Notes
Vehicle registration	KX19 LMN	AR19 XYZ	AR19XYZ	FAIL	The space is missing.
Country	France	United Kingdom	United Kingdon	FAIL	Updated, but misspelt.
Postcode	CF10 1AA	CF10 2BB	CF10 2BB	PASS	Applied as requested.

Overall: PARTIAL, 33% of changes completed, with a document reference, confidence score, tokens used and estimated cost. Values are illustrative.

Estimated effort

Activity	Days
Prerequisites: access, Azure OpenAI quota, Power Automate Premium licences, data policy approval for the custom connector, and agreeing the change reference format with the DH	0.5 – 1
Azure set-up: Azure OpenAI resource and model, function app, settings, publish and smoke test	0.5 – 1
Power Automate: custom connector and connection, email trigger, reading the reference, matching it to the change request, exception handling	1.5 – 2
Change audit records: results table in Dataverse, SharePoint or SQL with one row per change, outcome linked to the change request, human review queue for low-confidence results, notifications	1.5 – 2
Testing and validation: a range of real documents (PDF, RTF, Word, scans, multi-page and redacted), with prompt tuning and fixes	2
Handover: documentation and walkthrough	0.5
Total	**6.5 – 8.5 days**

Time spent waiting for IT approvals or quota increases is not included.

Running costs

- **Azure OpenAI:** pay per use. Typically a few US cents per check, and each result reports its own estimated cost.
- **Azure Functions:** consumption plan, likely to stay within the free monthly allowance at expected volumes.
- **Power Automate:** a Premium licence is required for the custom connector.

Risks and considerations

- The process depends on the DH quoting the change reference correctly; unmatched emails go to manual review.
- AI confidence is a self-assessment, so the threshold for human review should be set during testing.
- Exact-text checks need a live document with readable text, not a scanned image.
- Upkeep: .NET 8 support ends in November 2026 and GPT-4o retires in 2027; both need upgrading.